

MQL4GREC

Multimodal Quantitative Language for Generative Recommendation

A Paper Walkthrough: Bridging Domains & Modalities with a Unified Token Vocabulary

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Introduction: The Problem with Traditional Recommenders

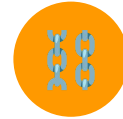
What is a Recommendation System? A system that predicts items (products, movies, songs) a user may like, used widely across e-commerce and social platforms.

For years, **IDRec** dominated: every item gets a unique, meaningless numeric ID (e.g. item_4521). Models learn purely from these IDs.



Cold-Start Problem

A brand-new item has an ID the model has never seen — so it cannot be recommended reliably.



No Knowledge Transfer

Random IDs carry no meaning, so what a model learns in one domain never helps another domain.

Recent generative approaches predict item-identifier tokens directly (seq2seq), leveraging pretrained language models (PLMs) like LLaMA/GPT — but they still ignore multimodal item content and struggle to transfer knowledge across domains.

Problem Statement

❖ PLM–RS Mismatch

General linguistic knowledge inside PLMs doesn't align with the specific requirements of recommendation tasks.

❖ Multimodal Info Ignored

Existing methods rarely exploit the complementary knowledge between an item's text and image content.

❖ Meaningless IDs

Random or sequential item IDs carry zero semantic signal — similar items look unrelated to the model.

❖ Domain Silos

Recommendation knowledge learned in one domain (e.g. Toys) cannot transfer to another (e.g. Instruments).

Goal: transform items from every domain and modality into ONE unified, meaningful token language.

Our Solution: MQL4GRec — An End-to-End System



Quantitative Translator

RQ-VAE converts item text & image content into shared token codewords



QLQ Tasks

NIG, AIG & QLA tasks enrich tokens with semantic + cross-modal knowledge



Pretrain + Fine-tune

T5-style encoder-decoder transfers knowledge from source to target domains

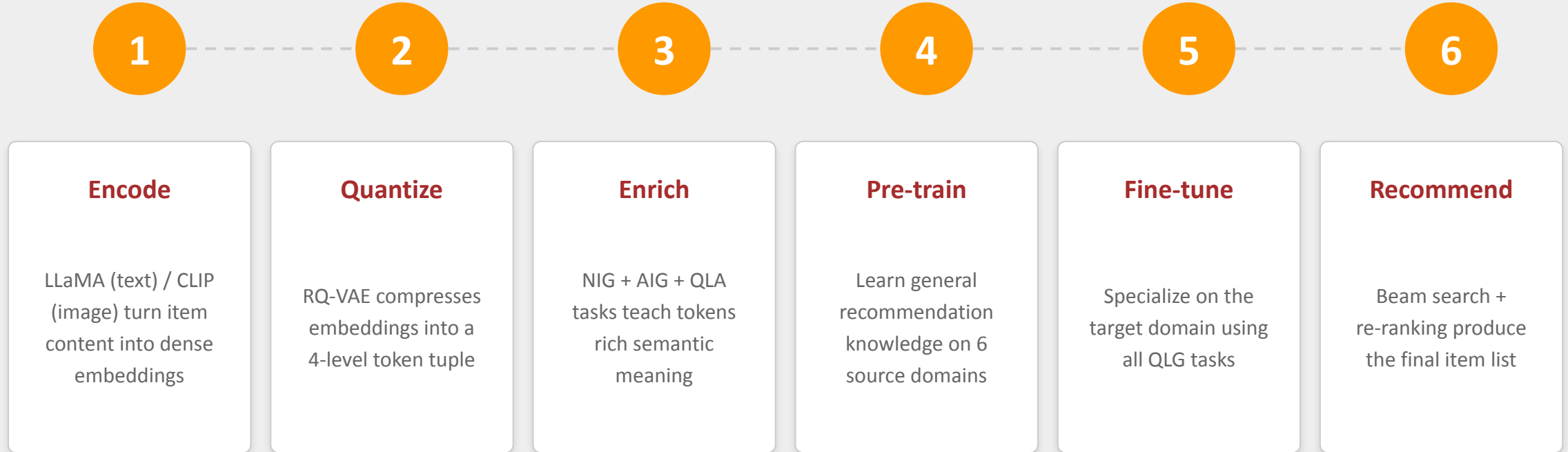


Re-ranking

Merges text- and image-based recommendation lists into one final ranking

★ Positioned as: items from different domains and modalities all share ONE vocabulary — this shared language becomes the bridge for transferring recommendation knowledge, not just an identifier.

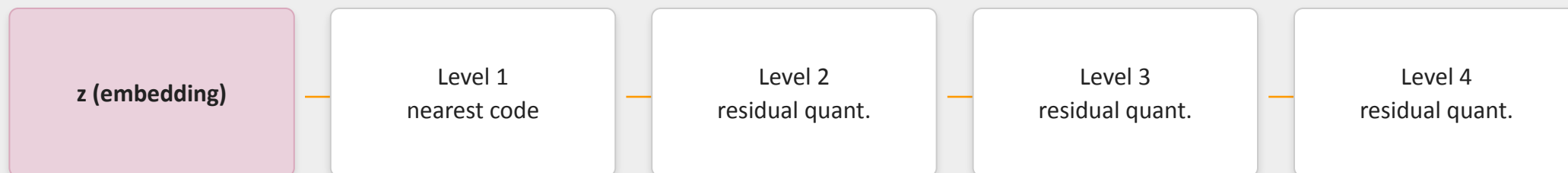
System Workflow: How It Works



★ *The tokens act as a universal recommendation "language" — just like a shared vocabulary lets NLP models transfer knowledge across tasks.*

The Quantitative Translator: Content → Tokens

RQ-VAE (Residual-Quantized VAE) turns one dense embedding into a tuple of discrete codewords by quantizing residuals, level after level.



Residual step: at each level, pick nearest codeword c_i , then residual $r_{i+1} = r_i - v(c_i)$ feeds the next level. Final tokens = (c_1, c_2, c_3, c_4) e.g. (2, 3, 1, 6).

Text Modality

Encoder: LLaMA

Codeword prefix: lowercase prefix

a_2 b_3 c_1 d_6

Image Modality

Encoder: CLIP ViT-L/14

Codeword prefix: UPPERCASE prefix

A_1 B_4 C_2 D_6

Training Objective: The Loss Functions

Two separate objectives are optimized — one to build the vocabulary (RQ-VAE), one to generate recommendations (Transformer).

① RQ-VAE Loss (builds the tokens)

$$L(h) = L_{\text{recon}} + L_{\text{rqvae}}$$

- $L_{\text{recon}} = \|h - \hat{h}\|^2$ — how well the decoded output $\hat{h}$ reconstructs the original item embedding h
- $L_{\text{rqvae}} = \sum \|sg[r_i] - v_i\|^2 + \beta \|r_i - sg[v_i]\|^2$ — pulls residuals and codewords together (sg = stop-gradient, β = weighting coefficient)

② Seq2Seq Generation Loss (drives recommendation)

$$L_{\theta} = - \sum_j \log P_{\theta}(Y_j \mid Y_{<j}, X)$$

- Standard negative log-likelihood — same loss style used to train language models like GPT
- X = encoder's input sequence (task prompt + source tokens)
- Y_j = the j -th target token being predicted, given everything generated so far

Token Vocabulary — A Worked Example

Item: *“Sengoku Basara: The Last Party”*

Encoded through the text / image quantitative translators, it becomes:

Text Quantitative Language

<a_2> <b_3> <c_1> <d_6>

Image Quantitative Language

<A_1> <B_4> <C_2> <D_6>

a / A = level-1 codebook • **d / D** = level-4 codebook • lowercase = text, UPPERCASE = image

Vocabulary size = 2LK (L levels × K codewords, × 2 modalities) → can uniquely represent up to K^L items, no manual IDs needed.

Handling Item Collisions

Problem: two different items can end up with the exact same token sequence — a “collision.”

✗ Prior Approaches (TIGER, P5)

- Append an extra identifier digit after the item tokens
- That extra digit is semantically unrelated — breaks the meaning of the language
- LC-Rec adds a uniform-distribution constraint, but still can't fully resolve collisions when content is identical

✓ MQL4GRec's Fix

- Measure distance between residual vectors and codeword vectors at every level
- Sort colliding items by their closest match at the last level
- Reassign the nearest available token first, working backward through levels if needed

Result: fewer forced/meaningless IDs, fairer evaluation, and no inflated metrics from unresolved collisions.

Quantitative Language Generation (QLG) Tasks

*_0 *_1 *_2

NIG

Next Item Generation

Main task. Given interaction history tokens, predict the next item's tokens — text→text (NIG1) or image→image (NIG2).

*_6 *_7 *_8

AIG

Asymmetric Item Generation

Cross-modal: image history → predict text tokens of next item (or vice versa).
Implicitly links the two modalities.

*_12 *_13 *_14

QLA

Quantitative Language Alignment

Direct alignment: given one item's text tokens, generate its image tokens (or vice versa) — no history involved.

★ *Special prompt tokens (*_0, *_1 ...) prepended to the input tell the model which task to perform — like a task instruction tag.*

Training: Pre-training → Fine-tuning

Two-stage paradigm, borrowed from NLP (pretrain-then-finetune) — a T5-style Transformer encoder–decoder does all the generation.

① Pre-training

- Source domains: Pet Supplies, Cell Phones, Automotive, Tools, Toys, Sports
- Task used: Next Item Generation (NIG) only
- Goal: learn general, transferable recommendation knowledge

② Fine-tuning

- Target domains: Musical Instruments, Arts & Crafts, Video Games
- Tasks used: ALL QLG tasks — NIG + AIG + QLA
- Goal: specialize + exploit multimodal user preferences

Architecture: 4-layer Transformer encoder + 4-layer decoder, 6 attention heads/layer · ~13M parameters · AdamW optimizer, batch size 4096 (pre-train) / 512 (fine-tune) · trained on Tesla V100 GPUs (4 for pre-training, 2 for fine-tuning).

Recommendation & Re-ranking

At inference, beam search generates two candidate lists — one per modality — which are then merged.

Text-based List (R_t)

From the Next Text Item Generation subtask, with per-item scores $s_t(x)$

Image-based List (R_v)

From the Next Image Item Generation subtask, with per-item scores $s_v(x)$

↓ merge ↓

Re-ranking rule:

Item in BOTH lists → $\text{score} = (s_t(x) + s_v(x)) / 2 + 1$ (boosted — both modalities agree)

Item in only ONE list → keep that list's original score

Intuition: items both modalities agree on are more likely to be genuinely relevant, so they're ranked higher.

Experimental Setup

Amazon Product Reviews dataset — 6 categories for pre-training (source), 3 for evaluation (target):

Role	Categories	Notes
Pre-training (Source)	Pet Supplies · Cell Phones & Accessories · Automotive · Tools & Home Improvement · Toys and Games · Sports & Outdoors	General recommendation knowledge only
Fine-tuning / Evaluation (Target)	Musical Instruments · Arts, Crafts & Sewing · Video Games	Full QLG tasks + final metrics reported here

Metrics: Recall@K and NDCG@K (K = 1, 5, 10), full-ranking (leave-one-out) evaluation · Beam size = 20 for generative methods.

Target Dataset	Users	Items	Interactions	Sparsity	Avg. Seq. Len
Instruments	17,112	6,250	136,226	99.87%	7.96
Arts	22,171	9,416	174,079	99.92%	7.85
Games	42,259	13,839	373,514	99.94%	8.84

Results: Performance vs. Baselines

NDCG improvement of MQL4GRec over the strongest baselines, as reported in the paper's abstract (average of NDCG@5 and NDCG@10 gains):

+11.18%

Instruments
(NDCG avg)

+14.82%

Arts
(NDCG avg)

+7.95%

Games
(NDCG avg)

Why MQL4GRec wins

- TIGER (text-only tokens) does well on Instruments/Arts but falls behind on Games due to no auxiliary content signal
- VIP5 (multimodal but PLM-based) underperforms — modality gap between PLMs and raw image features
- MQL4GRec unifies both modalities AND transfers cross-domain knowledge — best or near-best on every metric

Head-to-Head: MQL4GRec vs. TIGER (Actual Numbers)

TIGER is the strongest prior baseline (also RQ-VAE based). Raw scores from Table 1 of the paper:

Dataset	HR@10 (TIGER)	HR@10 (MQL4GRec)	NDCG@10 (TIGER)	NDCG@10 (MQL4GRec)
Instruments	0.1221	0.1375	0.0950	0.1060
Arts	0.1167	0.1327	0.0806	0.0950
Games	0.0857	0.1033	0.0453	0.0548

Reading the table:

- MQL4GRec beats TIGER on every metric, on all 3 datasets — no exceptions
- The gap is widest on Games (+20.5% HR@10, +21.0% NDCG@10) — exactly where TIGER's text-only tokens fall short
- Full paper Table 1 also compares against GRU4Rec, BERT4Rec, SASRec, FDSA, S³-Rec, VQ-Rec, MISSRec, P5-CID and VIP5 across HR@1/5/10 and NDCG@5/10

Ablation Study Insights

Collision Handling Works

Distance-based reallocation beats TIGER's extra-ID-digit approach on NDCG@10 across all 3 datasets, and on HR@10 for Instruments & Games (near-parity on Arts).

AIG Always Helps

Adding the Asymmetric Item Generation task consistently boosts performance — cross-modal signal matters.

QLA Needs AIG

QLA alone gives little benefit; paired with AIG it produces the best results — alignment needs cross-modal grounding first.

Pre-training: Mixed on Games

Helps Instruments & Arts steadily; on Games it can hurt — likely domain mismatch / overfitting with source data.

Bonus Finding: Zero-Shot Capability

Does a model pre-trained only on source domains say anything useful about a target domain it has NEVER been fine-tuned on?

# Source Domains Used to Pre-train	Instruments HR@10	Arts HR@10	Games HR@10
0 (random init)	0.00099	0.00113	0.00066
2	0.00240	0.00140	0.00109
4	0.00310	0.00298	0.00054
6	0.00345	0.00311	0.00116

What this means:

- With ZERO fine-tuning, more pre-training domains → consistently better (though still weak) scores on Instruments & Arts
- Games barely improves — the paper attributes this to scarce pre-training data and the model's limited parameter count, causing insufficient generalization

What Makes MQL4GRec Different

Method	Modality	Cross-domain Transfer	Item ID Strategy
TIGER	Text only	✗ No	RQ-VAE + extra digit for collisions
VIP5	Text + Image	✗ No	PLM prompt-based, weak image fusion
P5-CID	Text only	✗ No	Collaborative indexing
MQL4GRec	Text + Image	✓ Yes (6 source domains)	Unified quantitative language + smart collision fix

Gap this paper addresses:

No prior generative recommender combined a shared multimodal token vocabulary with explicit cross-domain pre-training AND a principled collision-resolution scheme — MQL4GRec is the first to unify all three.

Limitations & Conclusion

Limitations

- Slower inference — beam search + autoregressive generation vs. traditional ranking models
- Requires item content (text + image) — scenario of missing content not yet studied

Conclusion

- Unified quantitative language bridges domains + modalities without traditional item IDs
- Pre-training + fine-tuning transfers recommendation knowledge effectively
- A meaningful step toward universal, scalable recommendation models

★ *The system doesn't rely on traditional item IDs — it bridges domains and modalities through a shared, semantically-grounded token language, achieving 11.18% / 14.82% / 7.95% NDCG gains over baselines.*

Anticipated Questions

Q: Why RQ-VAE instead of simple hashing/clustering for IDs?

A: RQ-VAE preserves semantic distance — similar items land on similar codewords — whereas hashing gives every item an arbitrary, unrelated ID.

Q: Why does pre-training hurt the Games dataset?

A: Paper attributes it to overfitting / domain mismatch — Games differs more from the 6 source categories than Instruments or Arts do.

Q: How big is the model, and what hardware was used?

A: ~13M parameters; trained on Tesla V100 GPUs — 4 cards for pre-training, 2 for fine-tuning.

Q: What happens if an item has no image or no text?

A: Not addressed in the paper — explicitly listed as a limitation (needs item content for both modalities to work).

Q: Is inference slower than classic recommenders?

A: Yes — beam search + autoregressive generation is inherently slower than a single forward pass in traditional ranking models.

Q: How is collision handling actually validated?

A: Table 2 ablation: distance-based reallocation beats TIGER's extra-digit method on NDCG@10 in all 3 datasets, and HR@10 in 2 of 3 (near-parity on Arts).

References

Primary paper: Zhai, J., Mai, Z.-F., Wang, C.-D., Yang, F., Zheng, X., Li, H., & Tian, Y. (2025). Multimodal quantitative language for generative recommendation. In Proceedings of the International Conference on Learning Representations (ICLR 2025).

Rajput, S., Mehta, N., Singh, A., Hulikal Keshavan, R., Vu, T., Heldt, L., Hong, L., Tay, Y., Tran, V. Q., Samost, J., Kula, M., Chi, E. H., & Sathiamoorthy, M. (2023). Recommender systems with generative retrieval. In Advances in Neural Information Processing Systems 36 (NeurIPS 2023).

Geng, S., Tan, J., Liu, S., Fu, Z., & Zhang, Y. (2023). VIP5: Towards multimodal foundation models for recommendation. In Findings of the Association for Computational Linguistics: EMNLP 2023 (pp. 9606–9620). ACL.

Hua, W., Xu, S., Ge, Y., & Zhang, Y. (2023). How to index item IDs for recommendation foundation models. In Proceedings of the Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region (SIGIR-AP 2023) (pp. 195–204). ACM.

Zeghidour, N., Luebs, A., Omran, A., Skoglund, J., & Tagliasacchi, M. (2021). SoundStream: An end-to-end neural audio codec. IEEE/ACM Transactions on Audio, Speech, and Language Processing, 30, 495–507.

Zheng, B., Hou, Y., Lu, H., Chen, Y., Zhao, W. X., & Wen, J.-R. (2023). Adapting large language models by integrating collaborative semantics for recommendation. arXiv preprint arXiv:2311.09049.

Ni, J., Li, J., & McAuley, J. (2019). Justifying recommendations using distantly-labeled reviews and fine-grained aspects. In Proceedings of EMNLP-IJCNLP 2019 (pp. 188–197). ACL.

Thank You!

MQL4GRec shows that recommendation knowledge can be transferred across domains and modalities through one shared, meaningful token language — not brittle, arbitrary item IDs.

Questions & Discussion Welcome